

App Development Project Report

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1. Student Details

Name: Dheeraj

Roll Number: 24f2002344

Email: 24f2002344@ds.study.iitm.ac.in

About Me: I am IITM BS student with interest in development and data science.

2. Project Details

Project Title: Trekking management app

Problem Statement:

To design and build a web app that lets user, admin and staff perform actions based on their roles

Approach:

The app was built using Flask as the backend framework, for database SQLAlchemy is used, to solve this problem i have used role-based access.

3. AI/LLM Declaration

I used **ChatGPT (GPT-5)** to assist in writing for styling and layout and debugging. The extent of AI/LLM usage is around **8-10%**, limited to **code suggestions and documentation formatting**.

All final implementation logic and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
Chart.js	To visualize trends such as activity vs. time
Flask-Login	User authentication and session management
SQLite	Lightweight local database for storing user data

5. Database Schema / ER Diagram

Tables:

1. **User** — stores user account and role details (id, username, email, password_hash, role)
2. **Trek** — stores available expedition details (id, title, location, difficulty, duration, cost, available_slots)
3. **Booking** — logs user trek reservations and links users to treks (id, user_id, trek_id, payment_status, created_at)

Relationships:

- One-to-Many -> User - Booking
- One-to-Many -> Trek - Booking
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(Insert ER diagram image here – can be created using dbdiagram.io or draw.io or any other website)



7. Architecture and Features (optional)

Architecture Overview:

- **app.py** – main Flask application entry point
- **models.py** – database models and schema definitions using SQLAlchemy
- **/routes** – Flask Blueprints handling user authentication, admin controls, and booking routes
- **/templates** – Jinja2 HTML templates for frontend rendering

- /instance – directory storing the local SQLite database file
- requirements.txt – lists project dependencies (Flask, SQLAlchemy, etc.)

Key Features:

- **Role-Based Access Control** – Separate authentication, views, and dashboards for Users and Administrators.
 - **Trek Management** – Full CRUD (Create, Read, Update, Delete) functionality for administrators to manage expedition listings.
 - **Automated Booking System** – Allows users to reserve trek slots while automatically logging relationships in the database.
 - **Admin Data Visualization** – Real-time graphical metrics tracking user registrations and booking statistics.
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8. Video Presentation

Drive Link:

https://drive.google.com/file/d/1vpNNWrRAWg7AA7_6YvSbr7vtm4PssKk3/view?usp=drive_link

(Accessible to all with “View” permission.)